

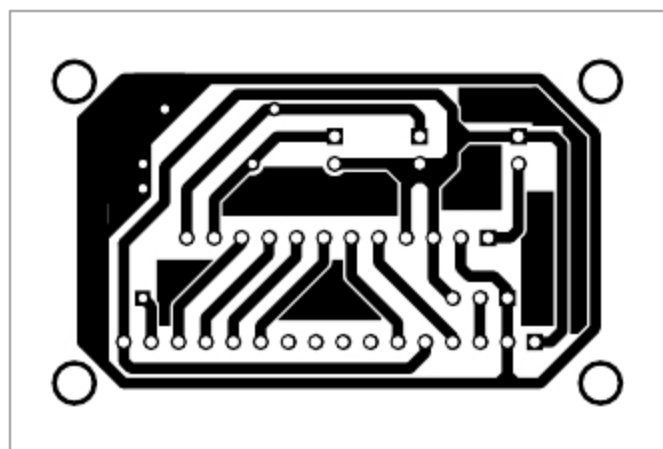


Fig. 3: Actual-size PCB layout for the timer kit

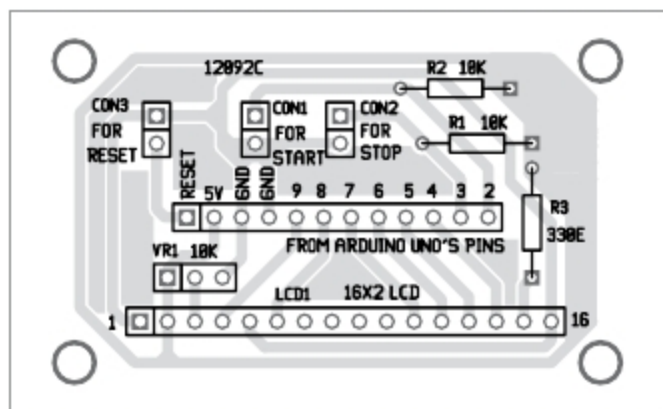


Fig. 4: Components layout of the PCB

Connect the Arduino Uno board and LCD1 as shown in the schematic diagram. Make sure that you first upload

the source code (timing\_kit.ino) to the Arduino Uno board via a USB cable attached to the laptop/desktop.

Connect start contacts to CON1, stop contacts to CON2, and reset push button to CON3. Power on the Arduino Uno board through USB/adaptor power supply. Adjust LCD1 contrast using potmeter VR1 to get a clear display.

Press the reset push button before carrying out time measurement. You will get the 'Timing test kit' message on the first line and 'Time(s): 0.000' on the second line of LCD1. The circuit is basically an up-counter/timer.

Now close the start contacts to start the count. The counted reading will be displayed on LCD1. Depending on your requirement, you can stop the count anytime by closing the stop contacts after a delay. The counting time is displayed in seconds on the left side and in fraction of seconds (in milliseconds) on the right side of LCD1,

which is the time duration between the start and stop of the timing test kit.

After assembling the circuit, enclose it in a suitable box with LCD1 display and reset button on the front panel. Start and stop connectors can be provided on the rear side of the panel for manual control or PLC connections. Now the kit is ready to use for measuring the timings of timer relays, circuit breakers, and similar devices.

**Note.** Only potential-free external N/O or N/C contacts shall be used as the start and stop contacts with a common ground. **EFY**

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